ARTIFICIAL INTELLIGENCE

CSE 3201

Constraint Satisfaction Problems

Constraint Satisfaction Problems

* eliminate large portions of the search space all at once

- **A CSP consists of:**
  - *Finite set of variables* $X_1, X_2, \ldots, X_n$
  - *Nonempty* ***domain*** *of possible values* for each variable $D_1, D_2, \ldots D_n$ *where* $D_i = \{v_1, \ldots, v_k\}$
  - *Finite set of constraints* $C_1, C_2, \ldots, C_m$
    - —Each *constraint* C_i limits the values that variables can take, e.g., $X_1 \neq X_2$.

    A *state* is defined as an *assignment* of values to some or all variables.

    A *consistent assignment* does not violate the constraints.
- **Example: Sudoku.**

Constraint Satisfaction Problems

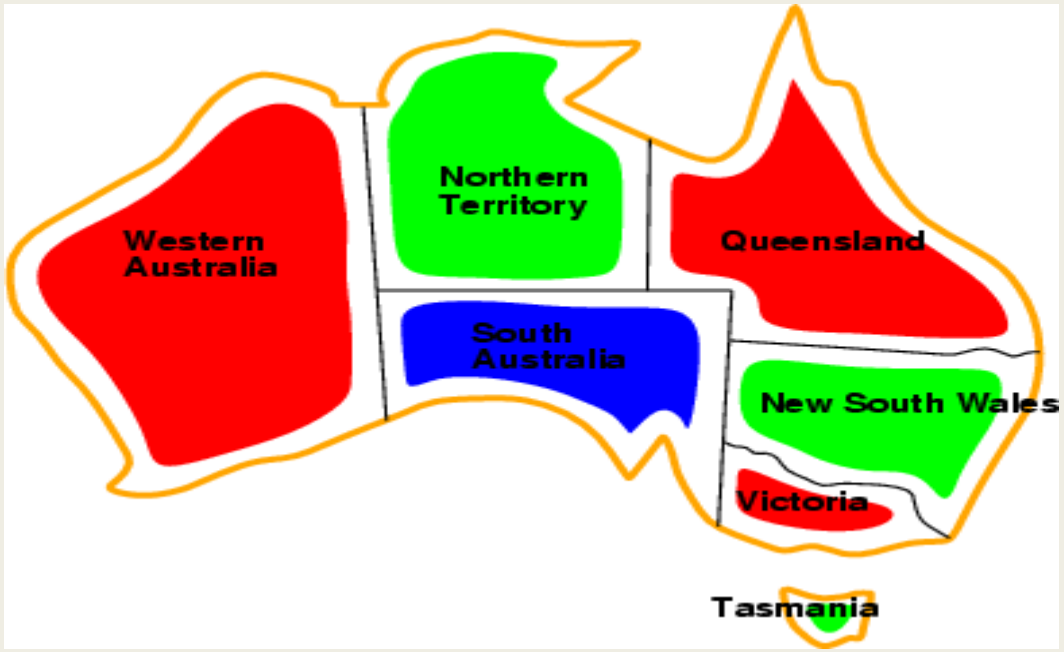
- An assignment is *complete* when every variable is assigned a value.
- A *solution* to a CSP is a complete assignment that satisfies all constraints.
- **Applications:**
  - Map coloring (coloring different regions of map, ensuring no adjacent regions have the same color)
  - Cryptarithmetic Problems (alphabets to numbers)
  - n-Queen (In an n-queen problem, n queens should be placed in an n x n matrix such that no queen shares the same row, column or diagonal.)
  - Sudoku (a number grid)
  - Crossword puzzle etc.

Example: Map-coloring



- **Variables:** ***WA, NT, Q, NSW, V, SA, T***
- **Domains:** D_i **= {red,green,blue}**
- **Constraints: adjacent regions must have different colors**
  - e.g., WA ≠ NT
    —So (WA,NT) must be in {(red,green),(red,blue),(green,red), …}

Example: Map-coloring



Solutions are complete and consistent assignments,

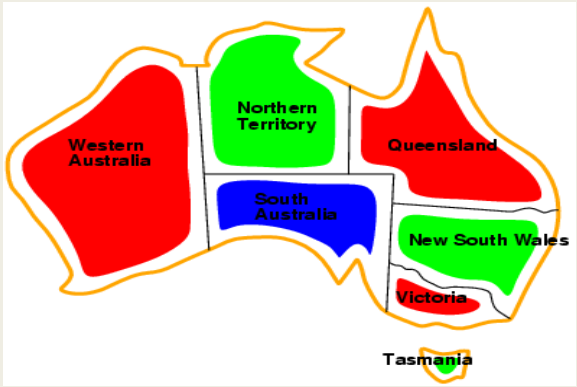
- e.g., WA = red, NT = green,Q = red,NSW = green,
  V = red,SA = blue,T = green

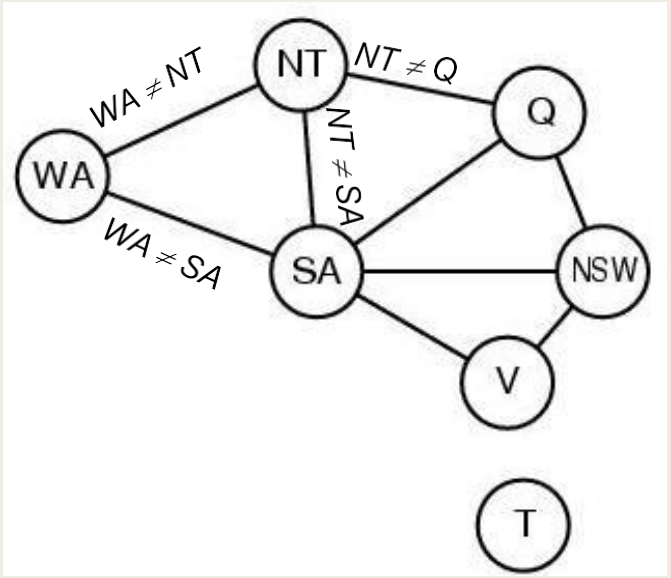
Benefits of CSP

- **Clean specification of many problems, generic goal, successor function**
  - Just represent problem as a CSP & solve with general package
- **CSP "knows" which variables violate a constraint**
  - And hence where to focus the search
- ***CSPs:* Automatically prune off all branches that violate constraints**
  - (State space search could do this only by *hand-building constraints into the successor function)*

CSP Representations

- ***Constraint graph:***
  - *nodes* are variables
  - *arcs* are constraints
- ***Standard representation pattern:***
  - variables with values
- ***This problem: A binary CSP:***
  - each constraint relates two variables





Varieties of constraints

- ***Unary*** **constraints involve a single variable,**
  - e.g., SA ≠ green
- ***Binary*** **constraints involve pairs of variables,**
  - e.g., SA ≠ WA
- ***Higher-order*** **constraints involve 3 or more variables**
  - e.g., crypt-arithmetic column constraints

CSP as a search problem

- **A CSP can easily be expressed as a search problem**
  - Initial State: the empty assignment {}.
  - Successor function: Assign value to any unassigned variable provided that there is not a constraint conflict.
  - Goal test: all the variables have assignments.
  - Path cost: a constant cost for every step.

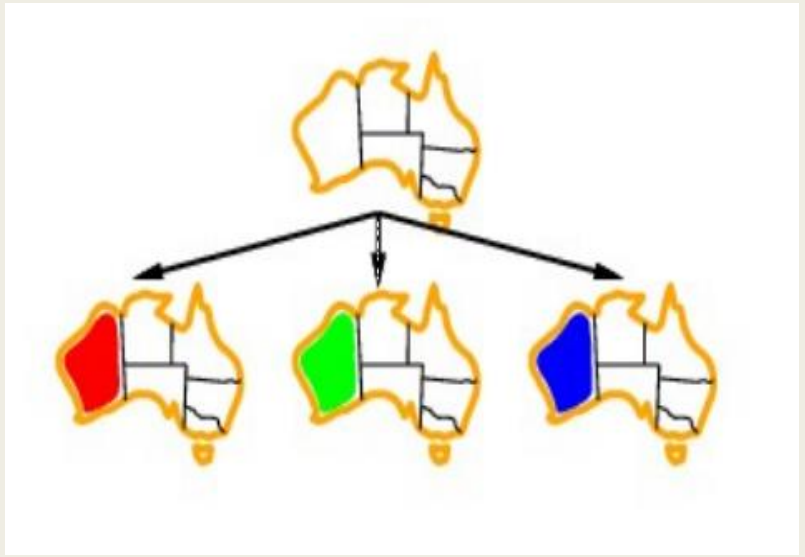
- **Solution is always found at depth *n*, for *n* variables**
  - Depth First Search can be used.

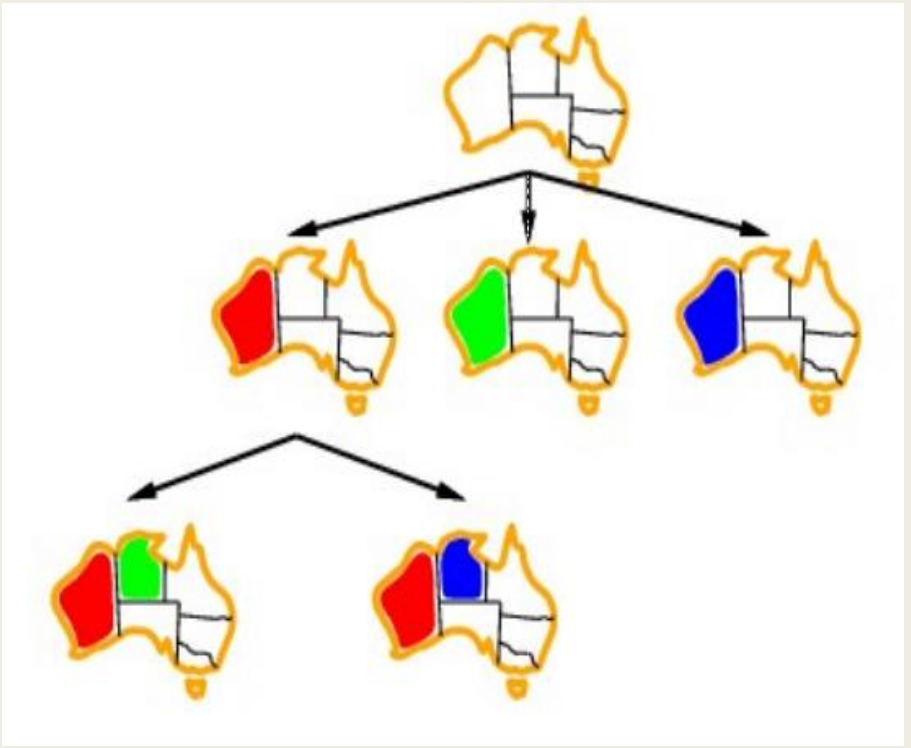
Backtracking search

- **Note that variable assignments are *commutative***
  - Eg *[step 1: **WA = red**; step 2: **NT = green**]*
    equivalent to *[step 1: **NT = green**; step 2: **WA = red**]*
- **Only need to consider assignments to a single variable at each node.**
- **Depth-first search for CSPs with single-variable assignments is called *backtracking* search.**
- **Backtracking search is the basic algorithm for CSPs.**

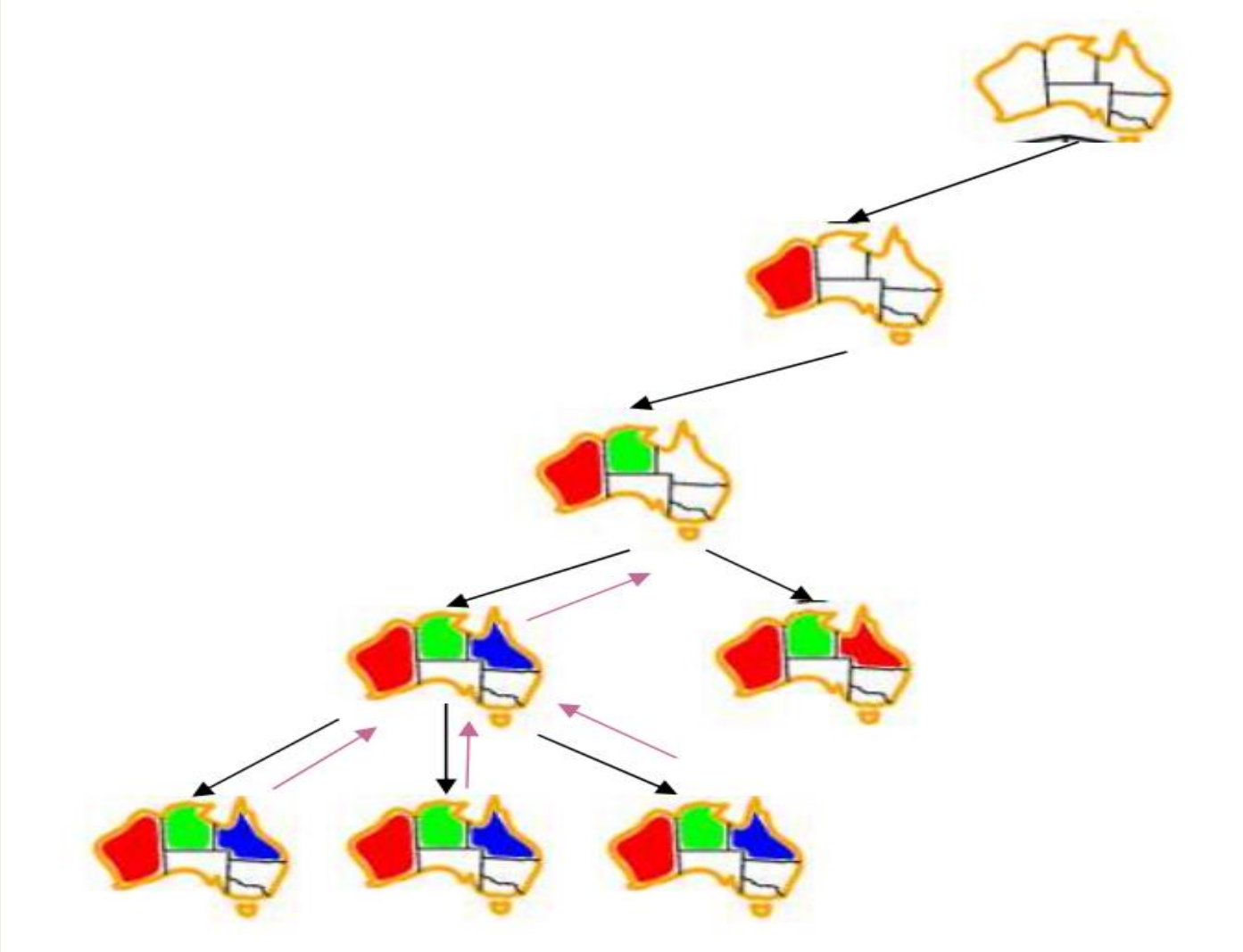
Example



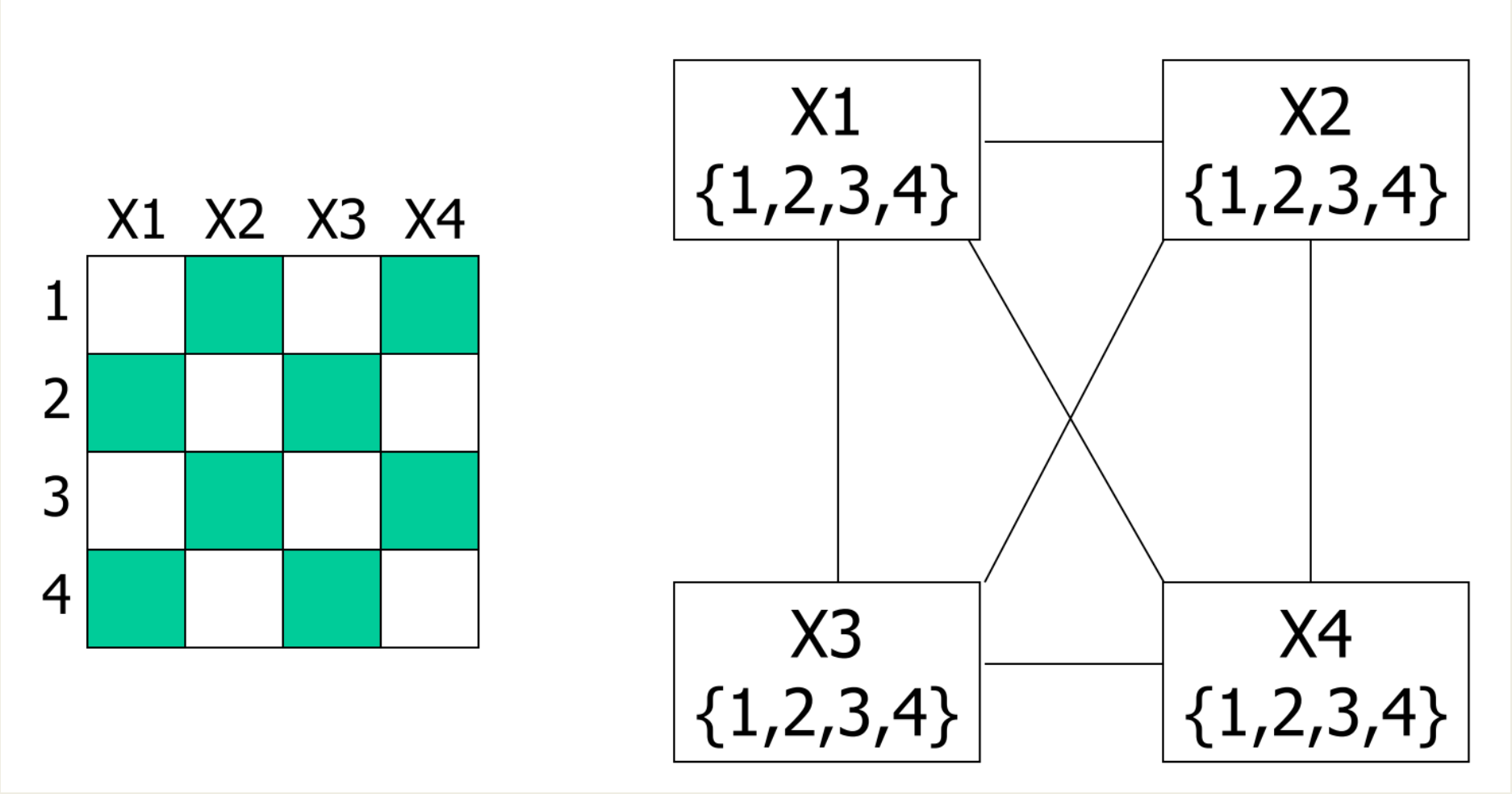




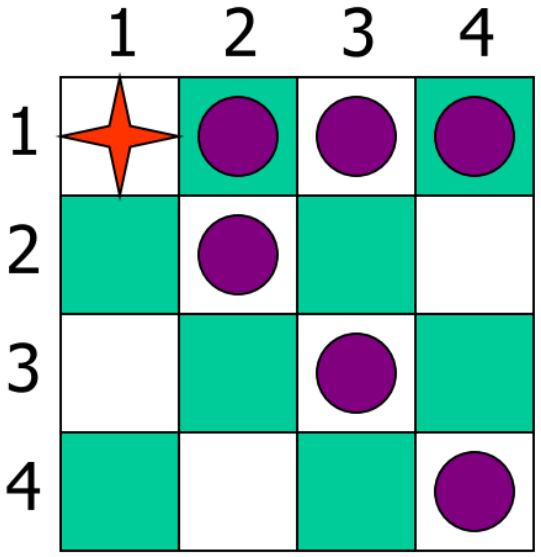
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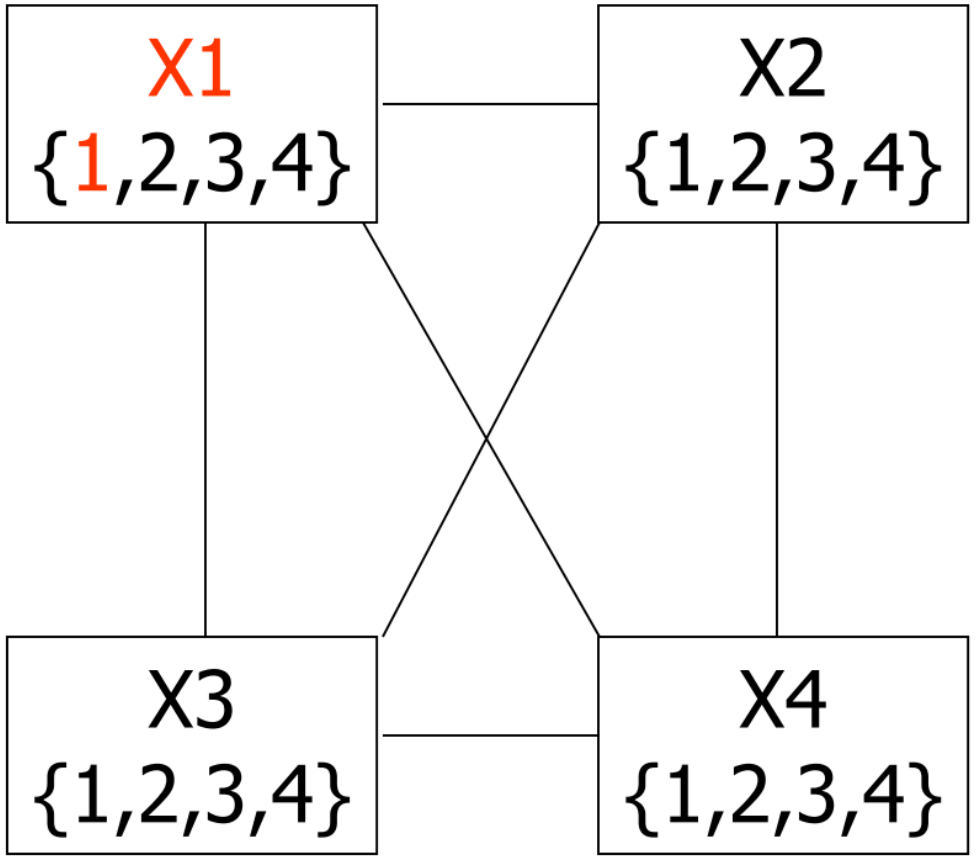


Example: 4-Queens Problem

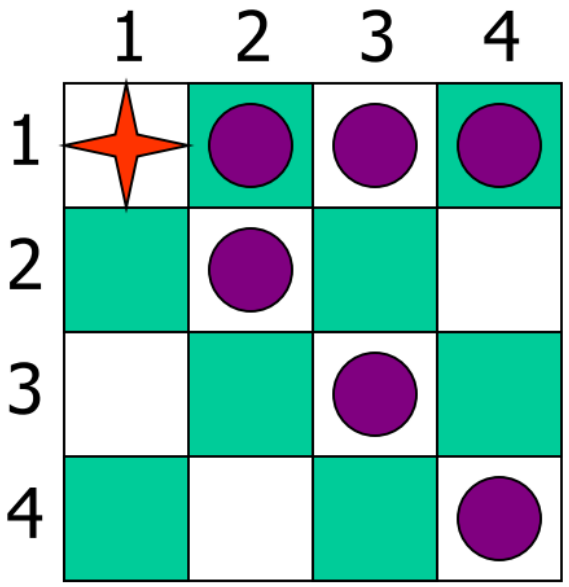


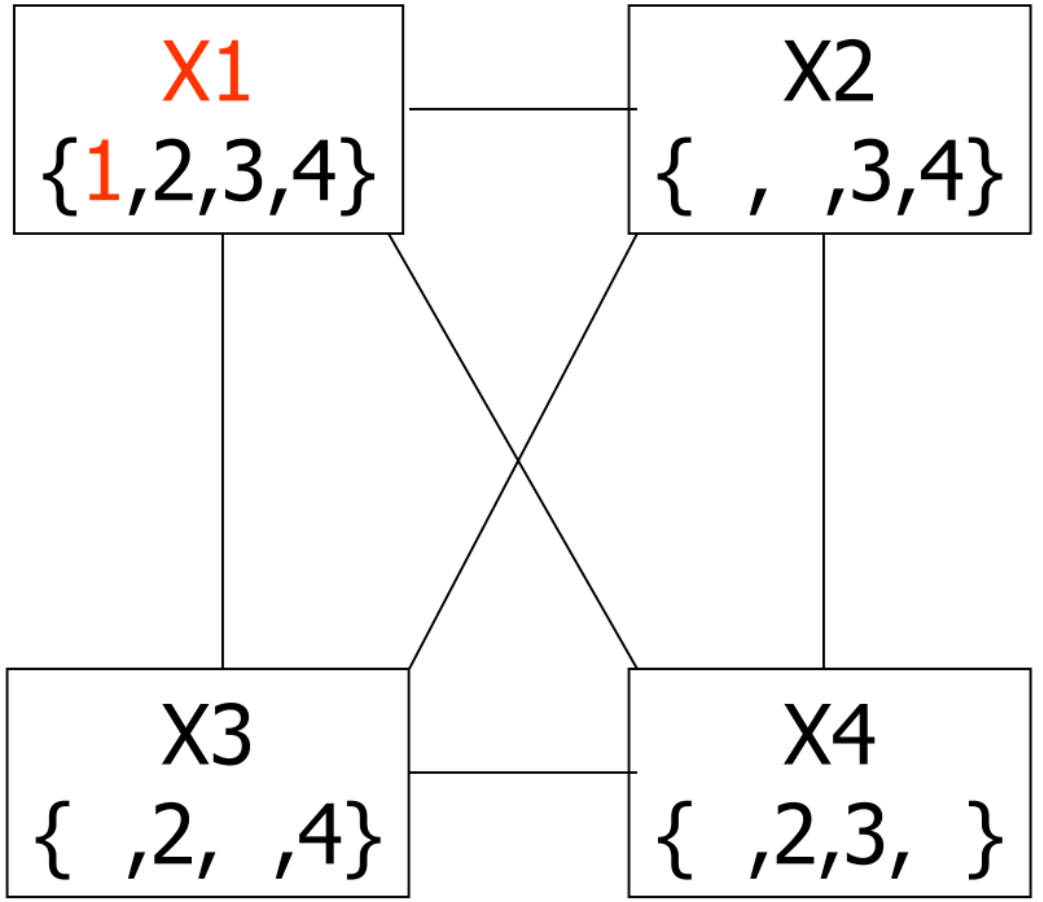
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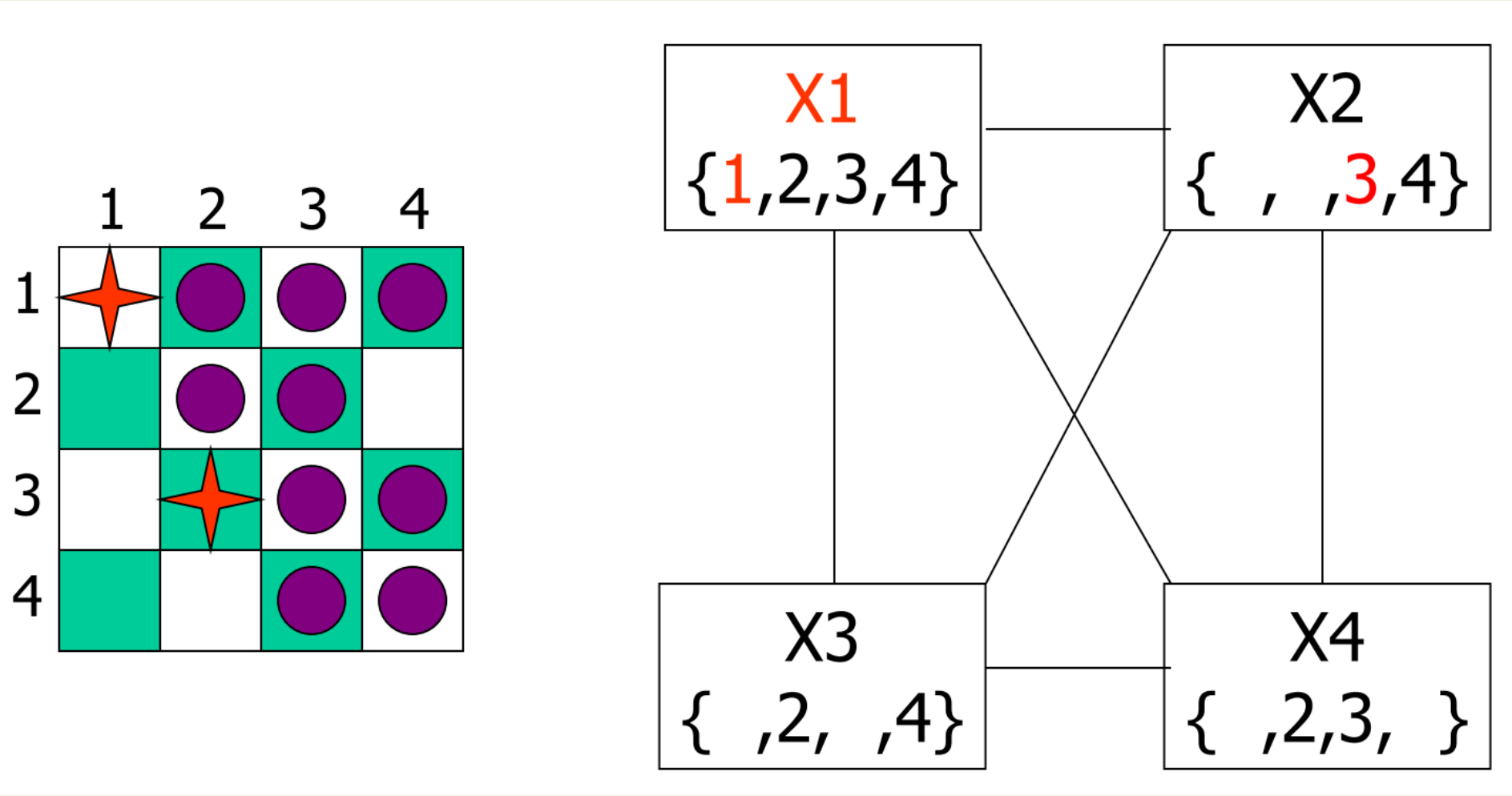


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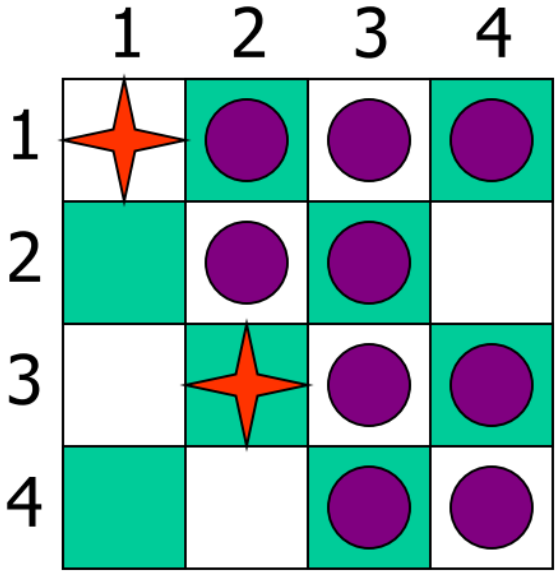


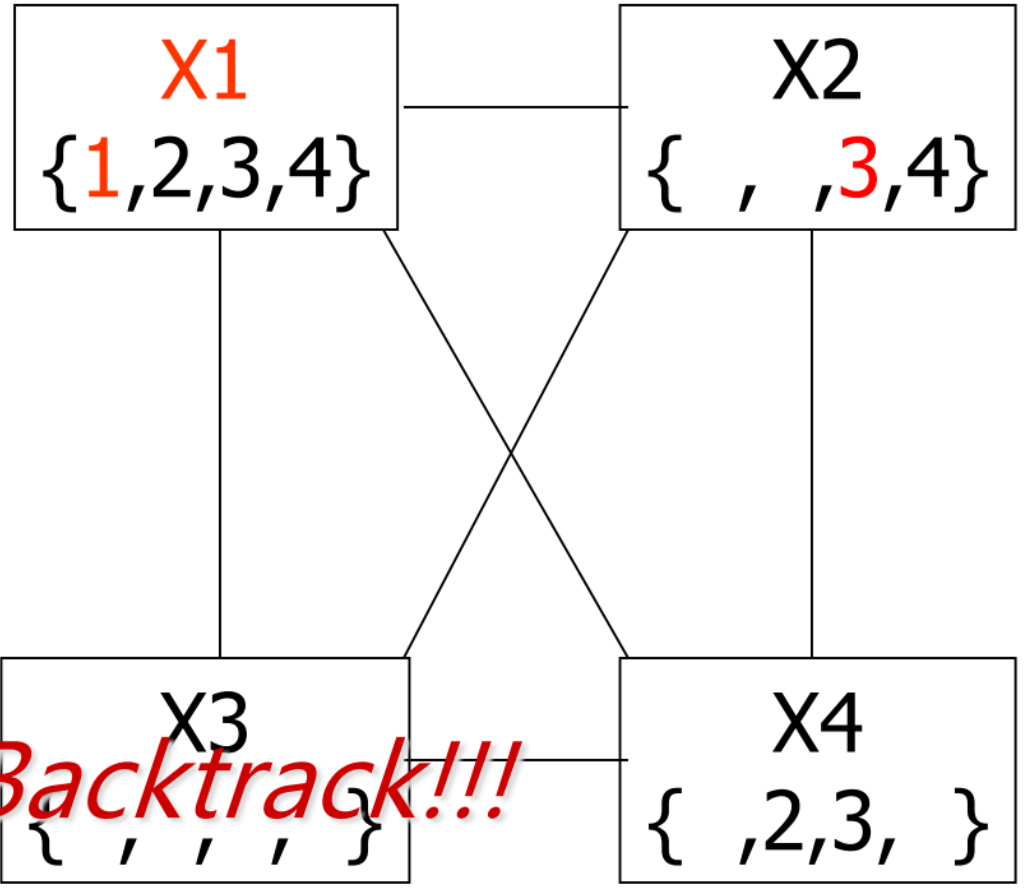


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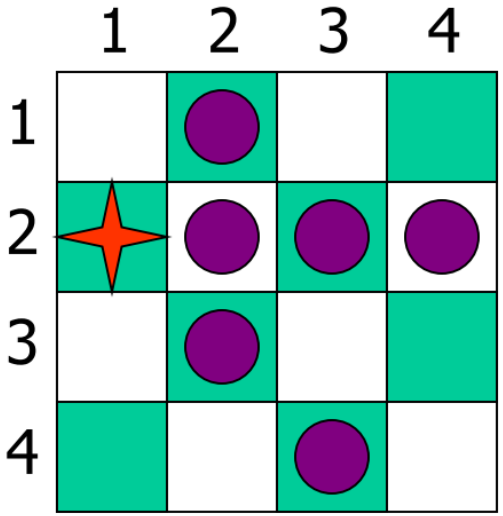
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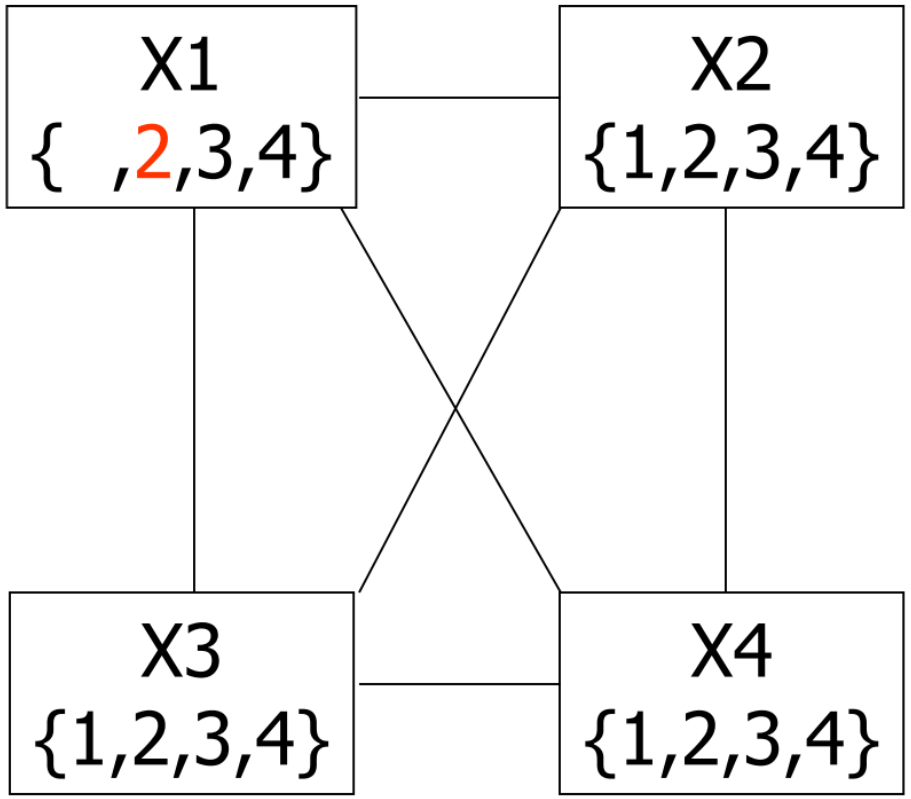




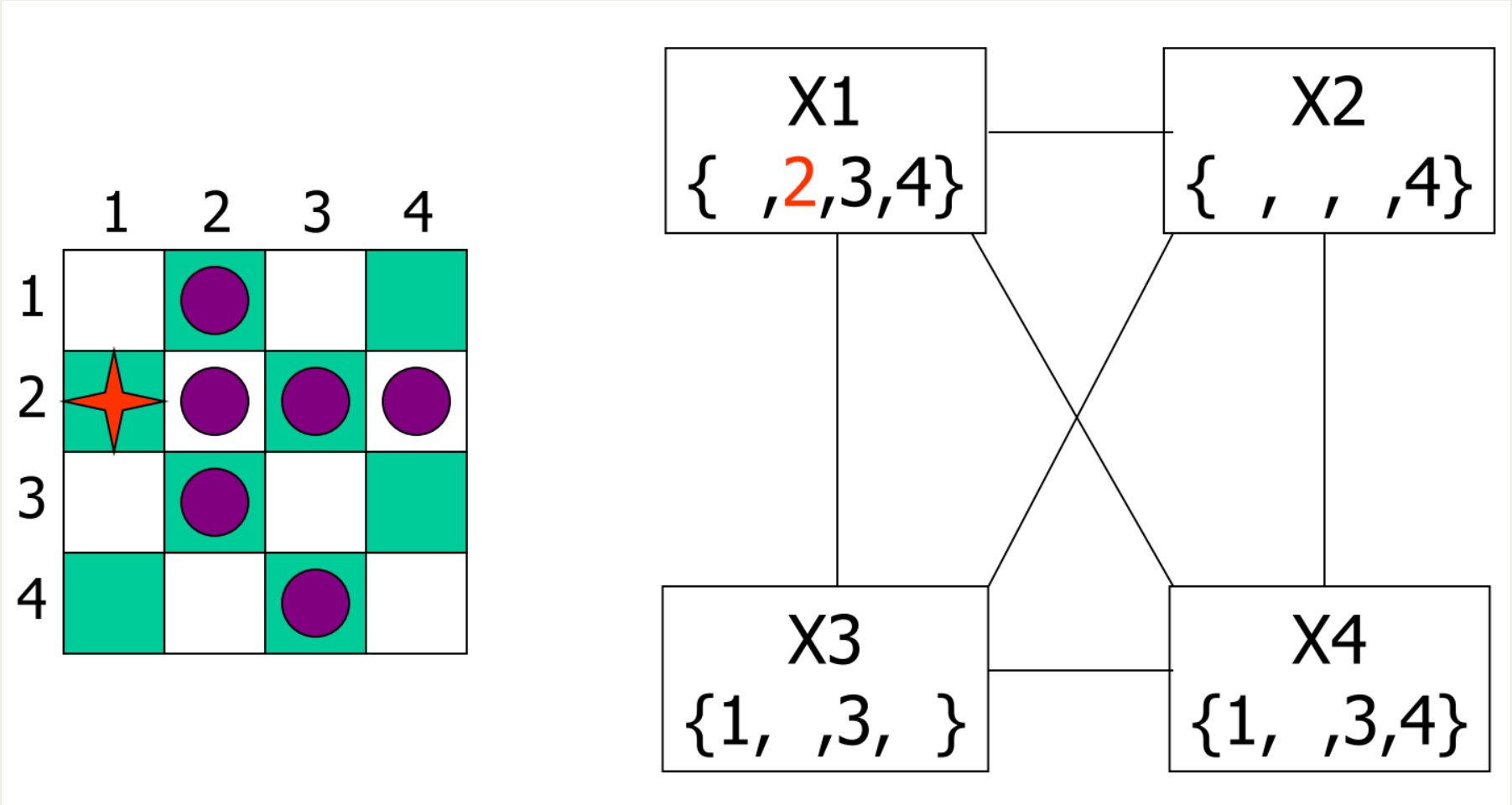
Example: 4-Queens Problem

Picking up a little later after two steps of backtracking….

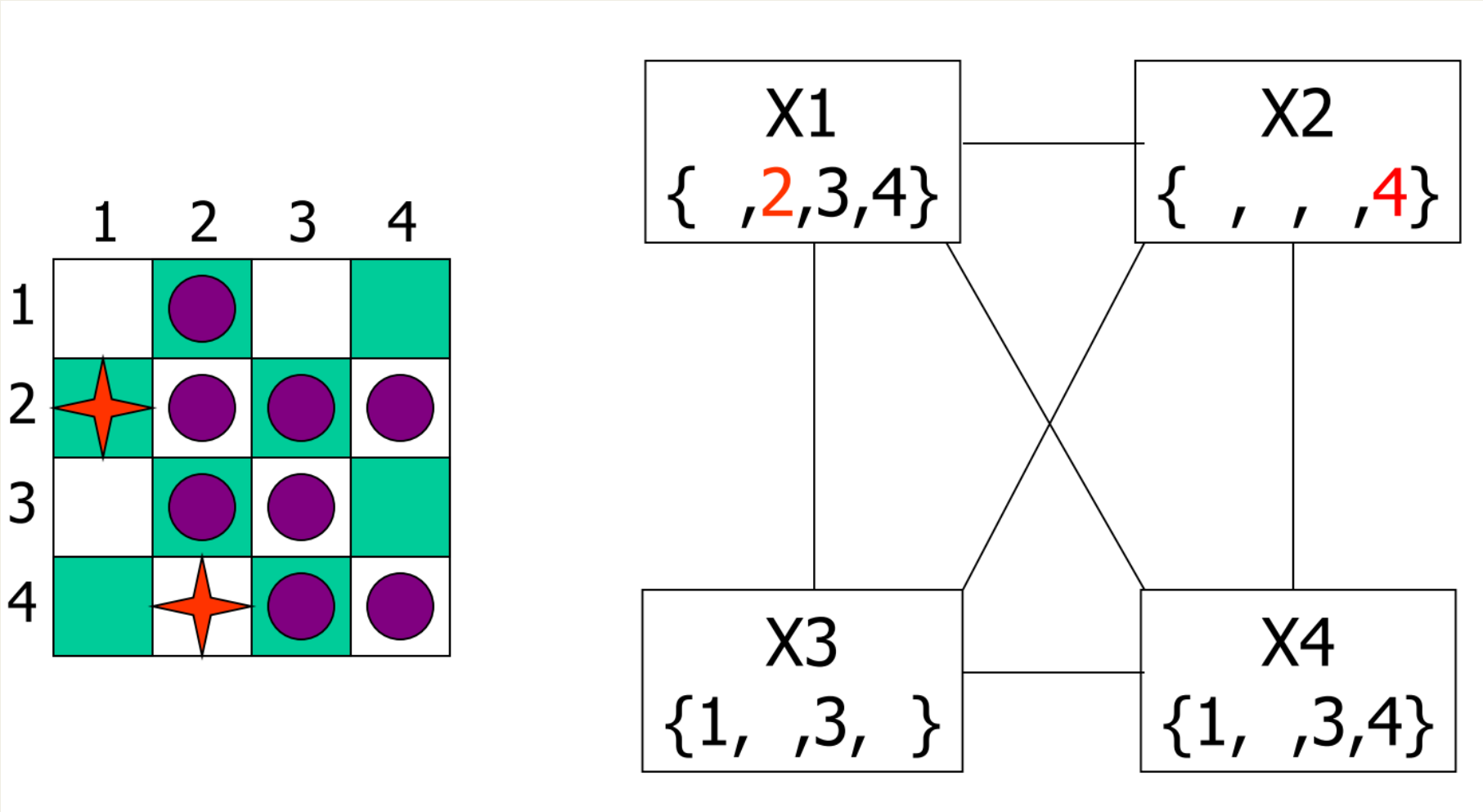




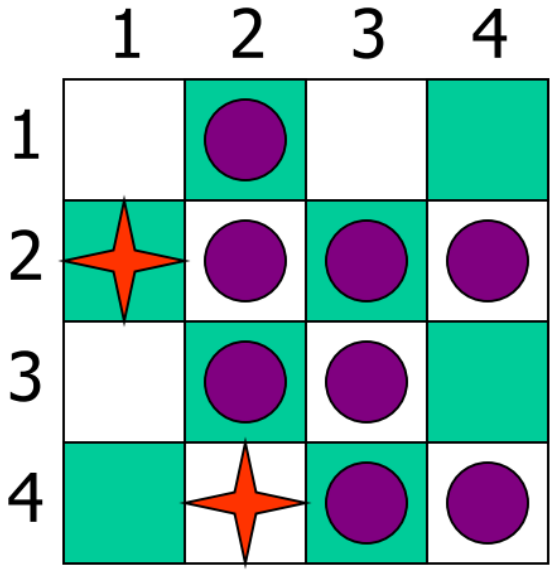
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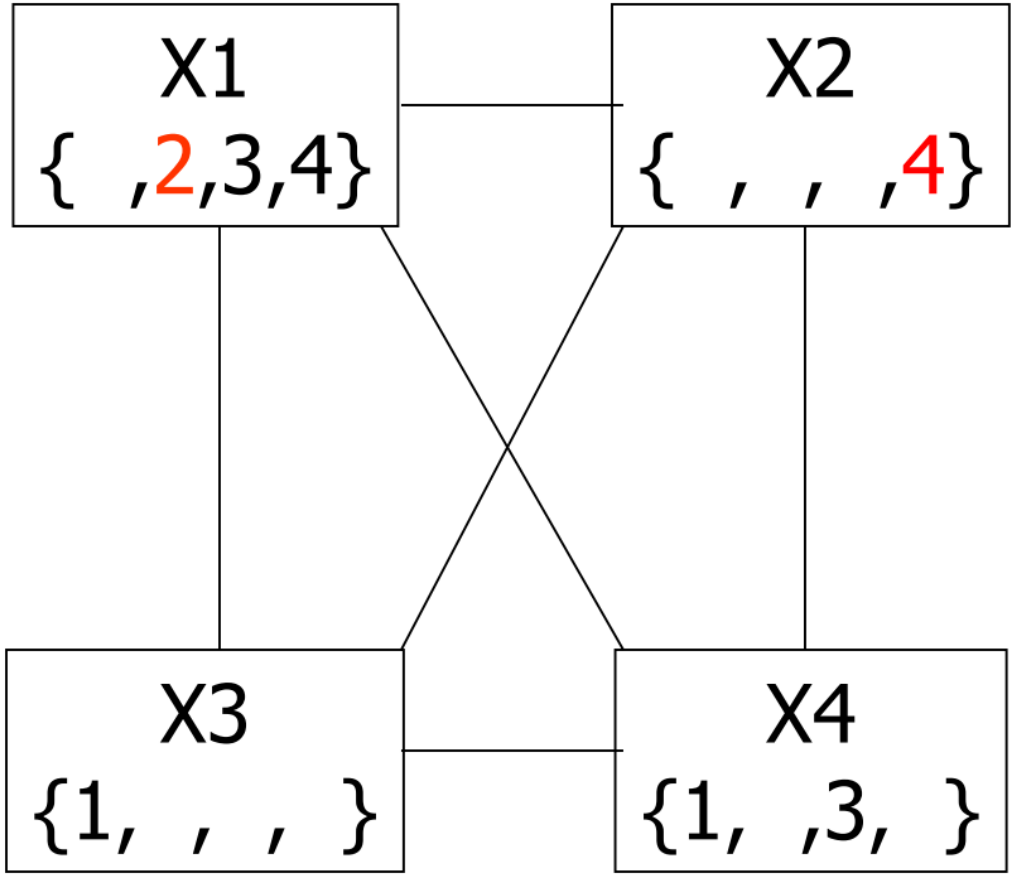


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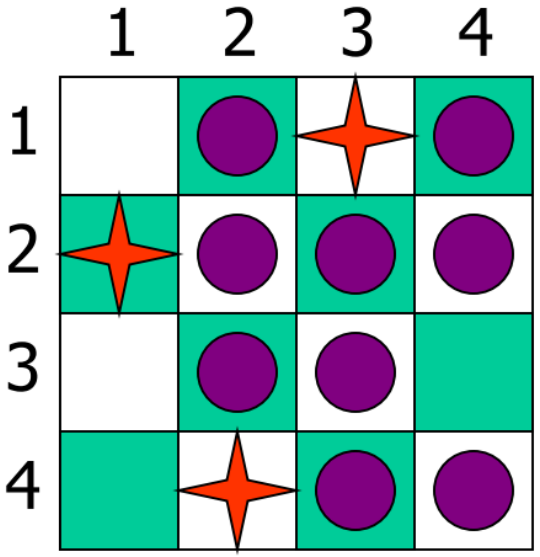


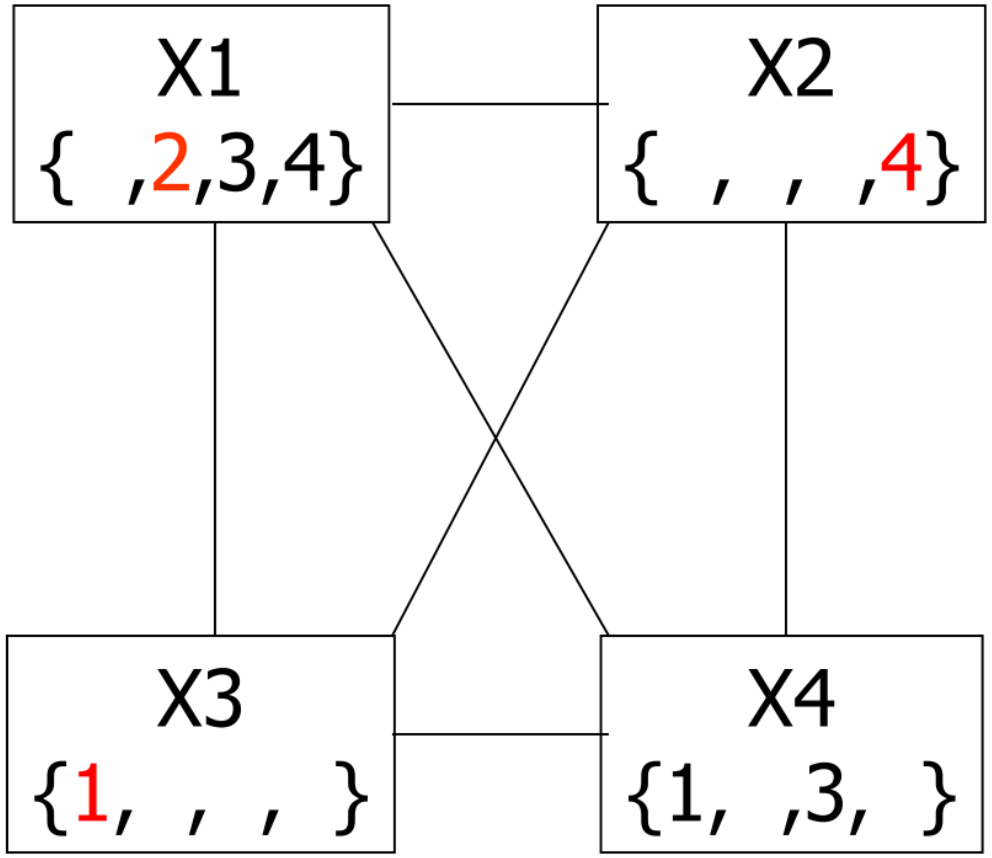
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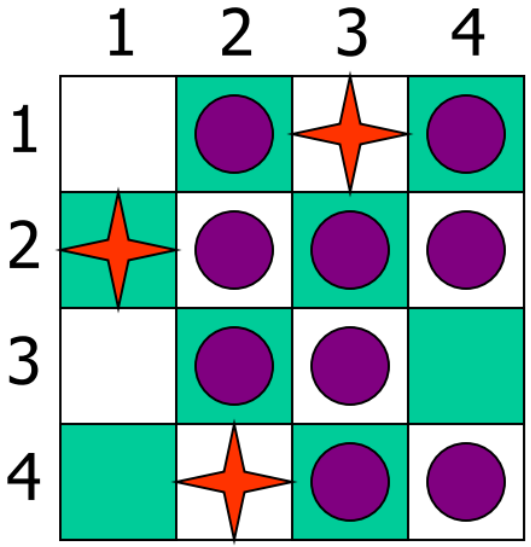


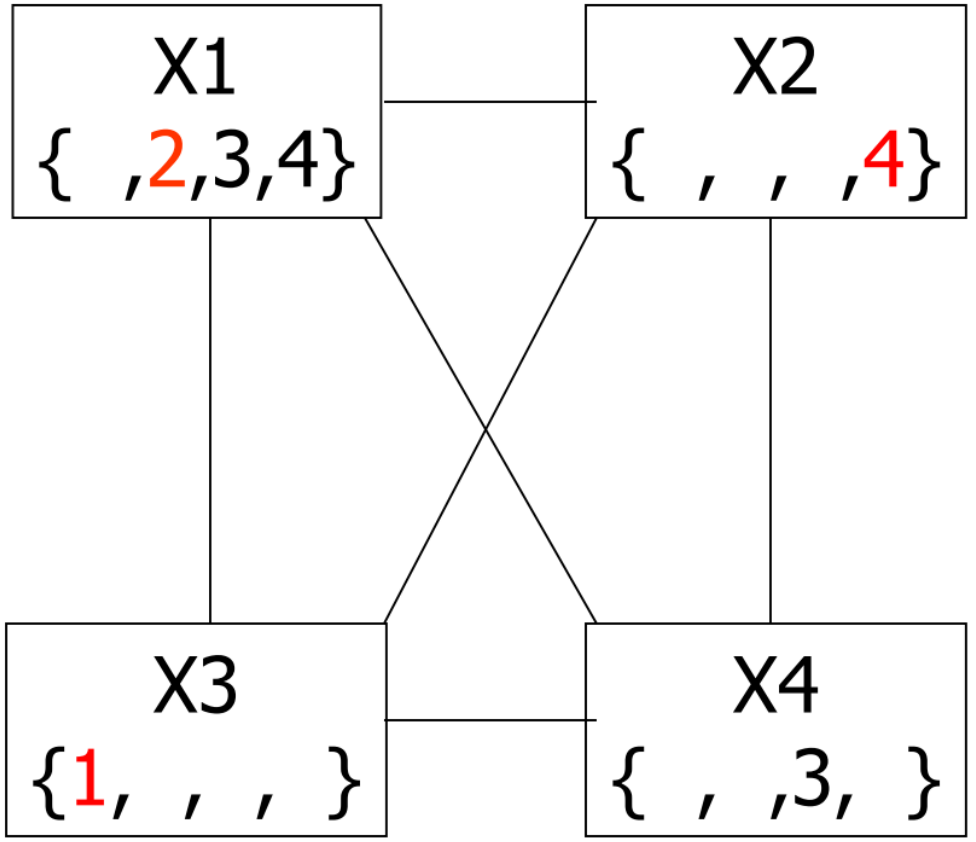
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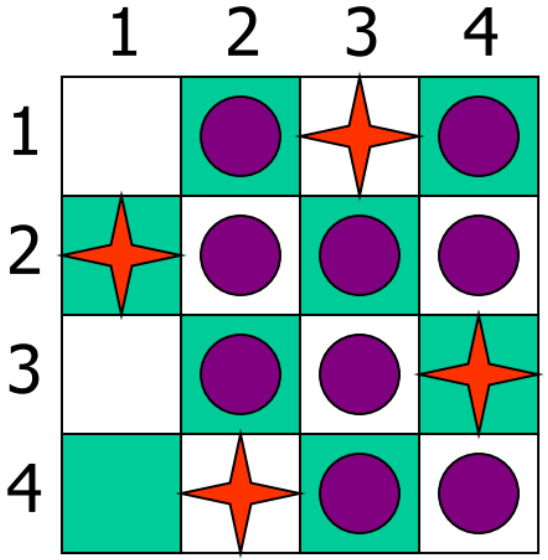


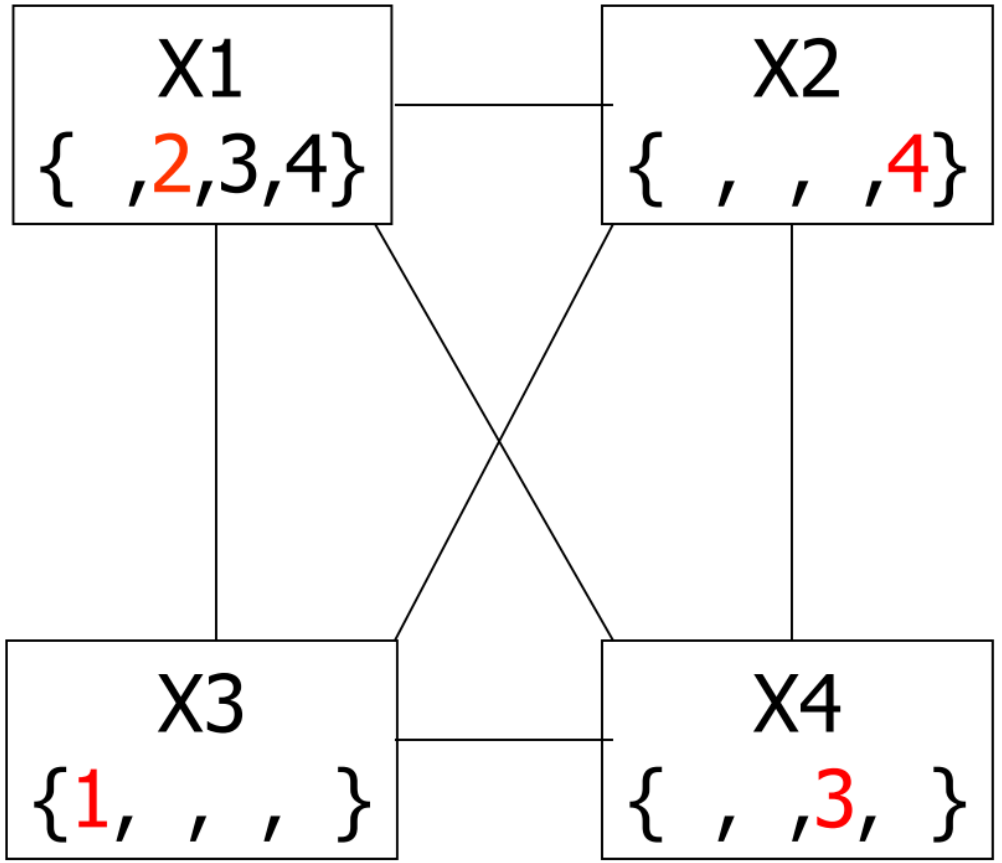
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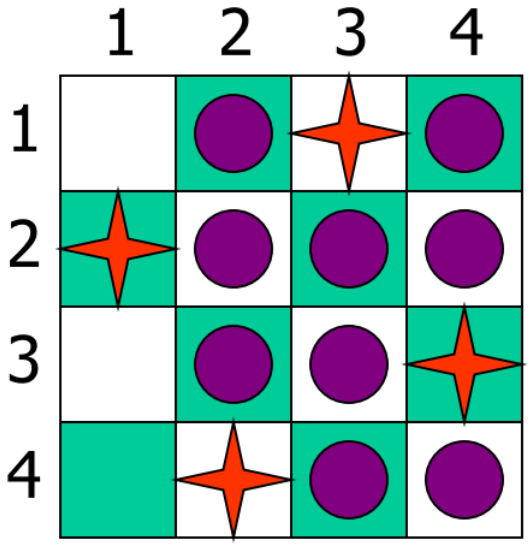


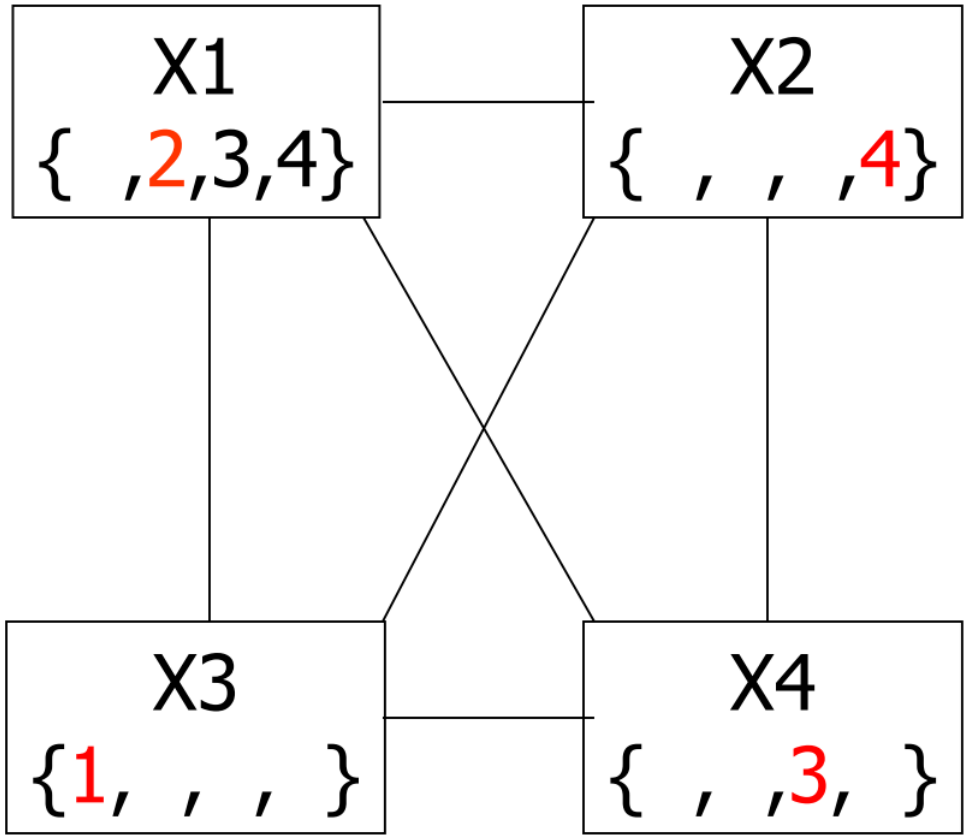
Example: 4-Queens Problem





Example: 4-Queens Problem





Cryptarithmetic Problem

- Cryptarithmetic Problem is a type of constraint satisfaction problem where the game is about digits and its unique replacement either with alphabets or other symbols. In **cryptarithmetic problem,** the digits (0-9) get substituted by some possible alphabets or symbols. The task in cryptarithmetic problem is to substitute each digit with an alphabet to get the result arithmetically correct.

Cryptarithmetic Problem Rules

1. Digit range should be **0 to 9**
2. Each letter represents **only one digit** throughout the problem and each variable should have **unique value**.
3. Numbers **must not begin with zero** (0134 is wrong, 1043 is correct)
4. You have to find the value of each letter in the problem.
5. After replacing letters by their corresponding digits, the result of the arithmetic operation must be correct.

Cryptarithmetic Problem

$$\begin{array}{r} \text{SEND} \\ +\ \text{MORE} \\ \hline \text{MONEY} \end{array}$$

- We have every letters standing for a digit and every letter stands for a different digit.
- We have to find an assignment of letters to digits such that a given arithmetic formula is correct.
- Variables are D, E, M, N, O, R, S, Y
- Domains are
  - {0, 1,2,3,4,5,6,7,8,9} for D, E,N,O,R,Y
  - {1,2,3,4,5,6,7,8,9} for S, M

Cryptarithmetic Problem

Constraint 1:

We can write one long constraint for the sum.

1000* S + 100* E+10* N+ D

+ 1000* M+ 100* O+10* R+ E

.....

10000*M + 1000* O+ 100*N+ 10* E+ Y

Constraint 2:

alldifferentr(D, E, M, N, O, R, S, Y)

These two constraints express the problem precisely.

Cryptarithmetic Problem

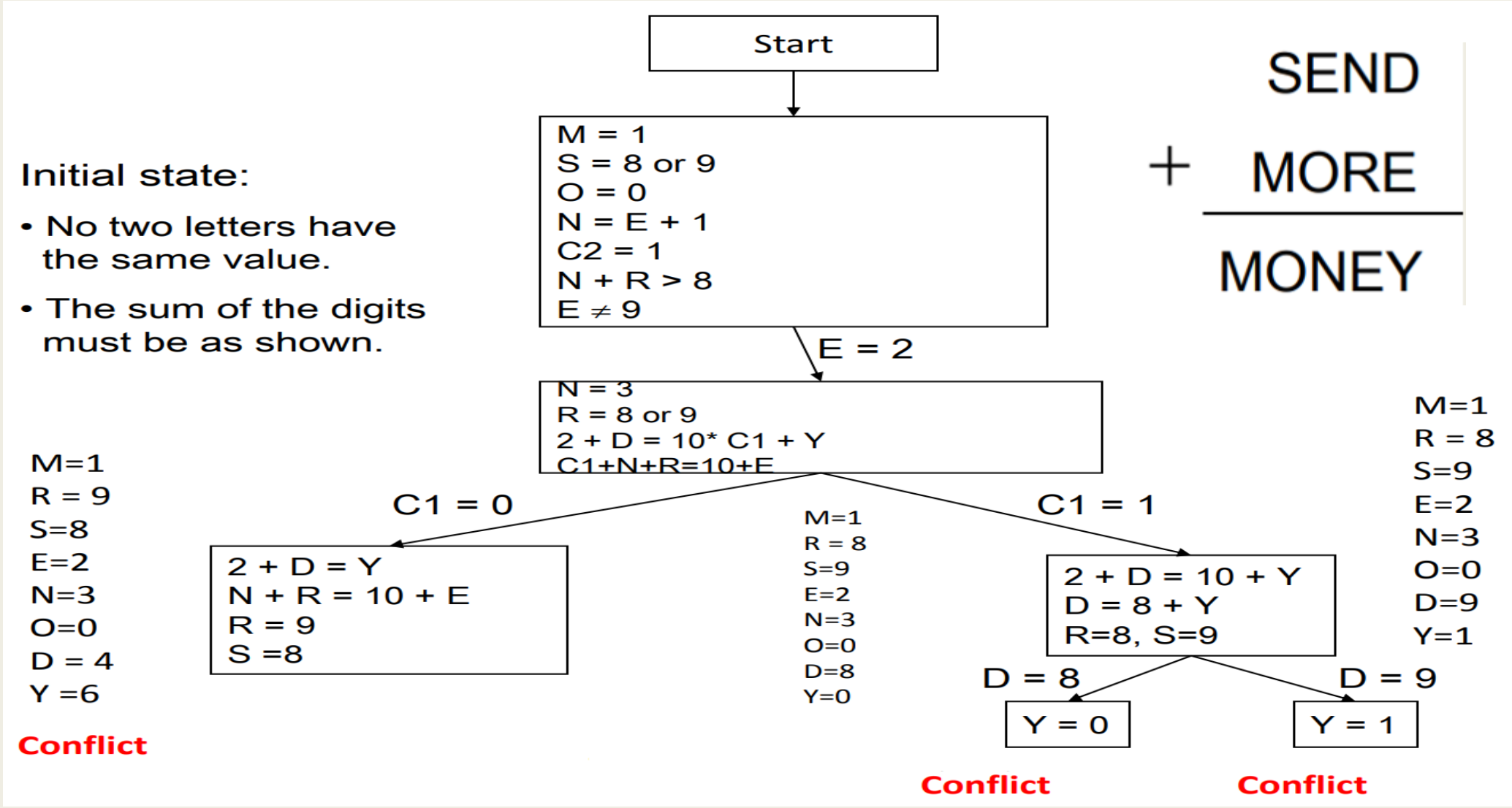
- **Rules for propagating constraints generates the following constraints:**

1. M = 1, since two single-digit numbers plus a carry can not total more than 19.
2. S = 8 or 9, since S+M+C3 > 9 (to generate the carry) and M = 1, S+1+C3>9, so S+C3 >8 and C3 is at most 1.
3. O = 0, since S + M(1) + C3(<=1) must be at least 10 to generate a carry and it can be most 11. But M is already 1, so O must be 0.
4. N = E or N=E+1, depending on the value of C2. But N cannot have the same value as E. So N = E+1 and C2 is 1.
5. In order for C2 to be 1, the sum of N + R + C1 must be greater than 9, so N + R must be greater than 8.
6. N + R cannot be greater than 18, even with a carry in so E cannot be 9.

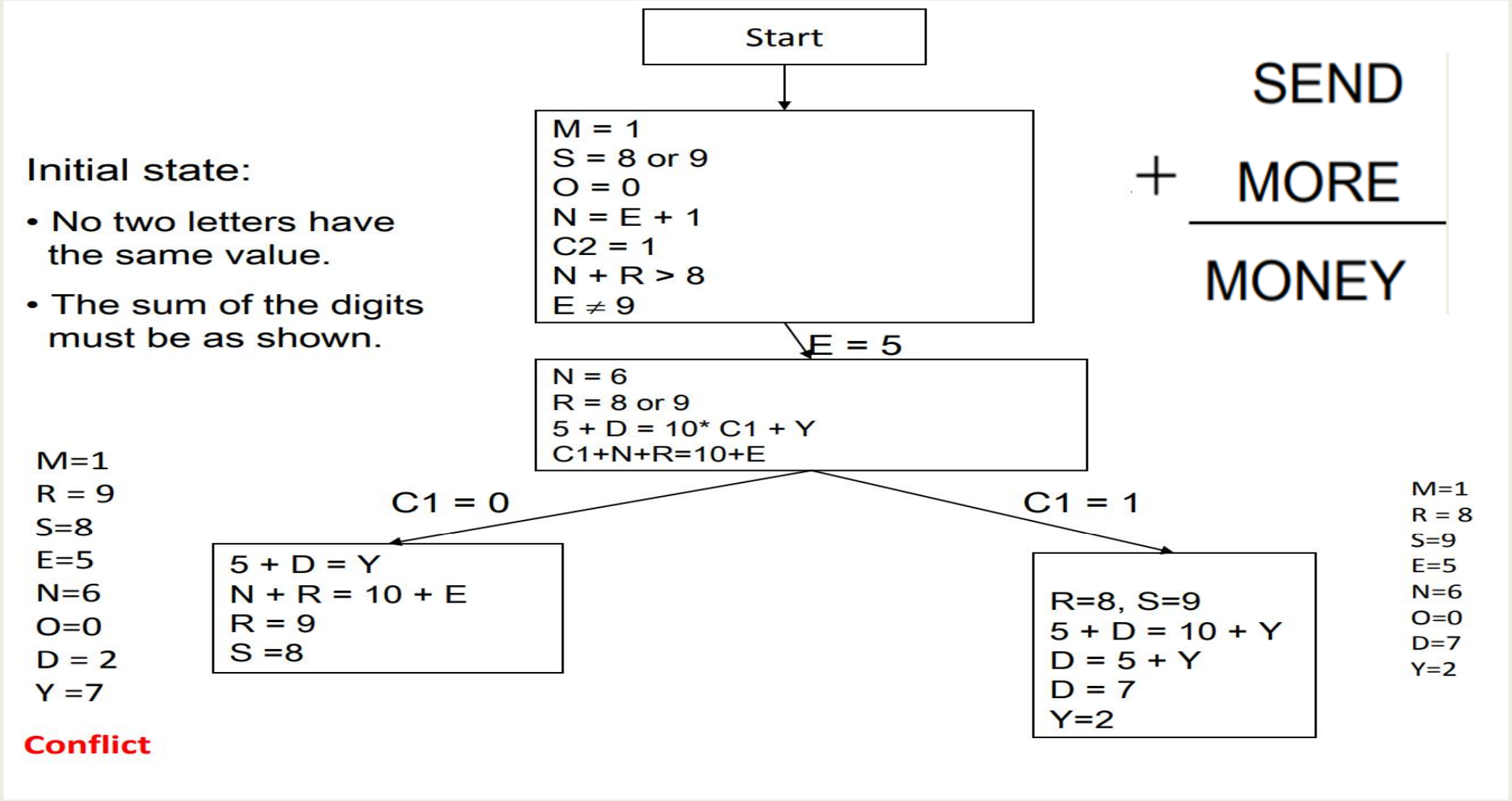
Cryptarithmetic Problem

- Suppose E is assigned the value 2.
- The constraint propagator now observes that:
- N = 3 since N = E + 1.
- R = 8 or 9, since R + N (3) + C1 (1 or 0) = 2 or 12. But since N is already 3, the sum of these nonnegative numbers cannot be less than 3. Thus R + 3 +(0 or 1) = 12 and R = 8 or 9.
- 2 + D = Y or 2 + D = 10 + Y, fro the sum in rithmost column.

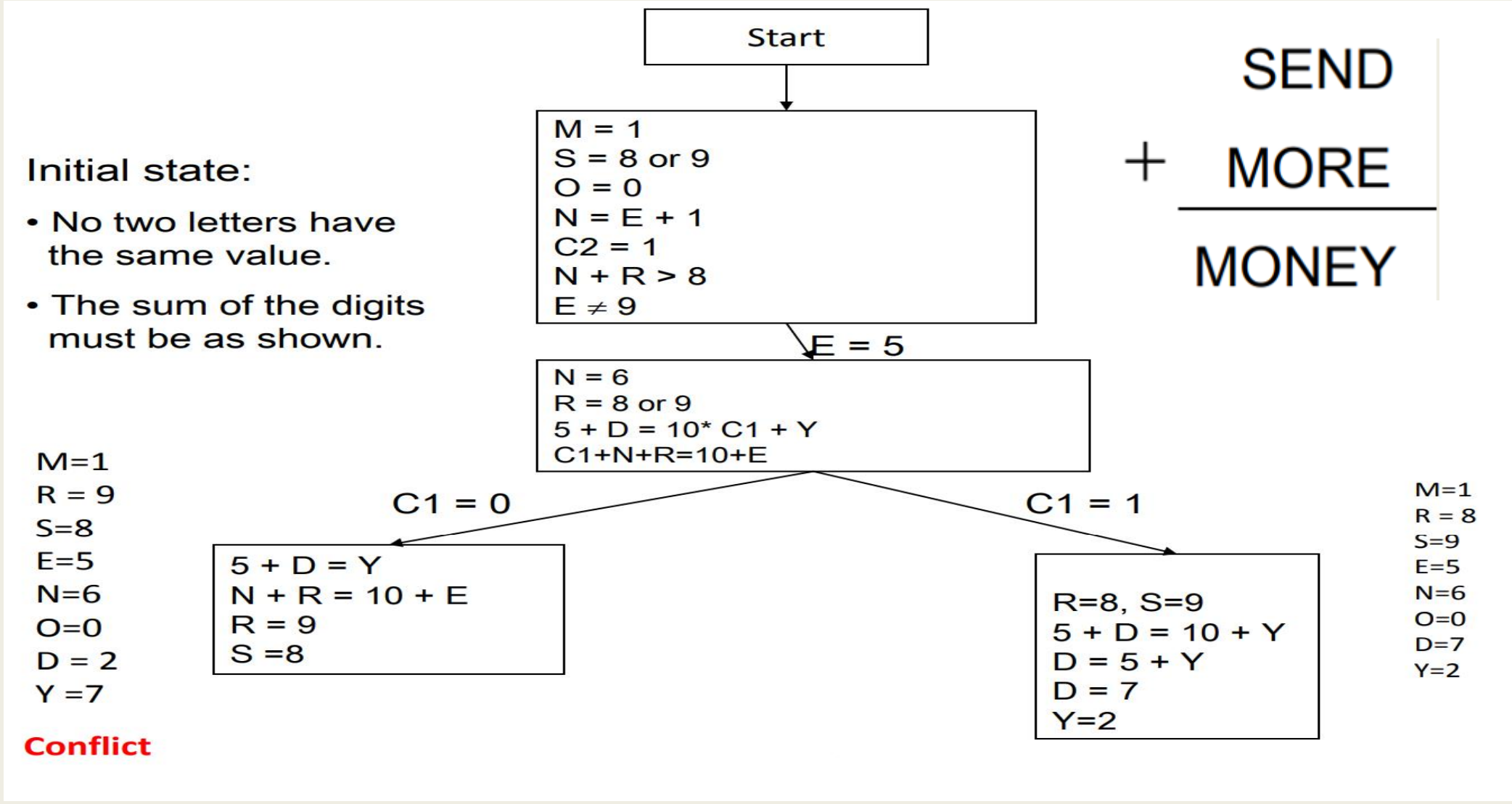
Tracing cryptarithmetic problem



Tracing cryptarithmetic problem



Tracing cryptarithmetic problem



Solution

- Keeping all the constraints in mind, the final resultant is as follows:

	S	E	N	D
+	M	O	R	E
M	O	N	E	Y

```
   9567
+  1085
--------
  10652
```

S	9
E	5
N	6
D	7
M	1
O	0
R	8
y	2

Exercise

$$\begin{array}{r} \text{CROSS} \\ +\text{ROADS} \\ \hline \text{DANGER} \end{array}$$

Variables : $X = \{C, R, O, S, A, N, D, G, E\}$.
Each variable has the domain $\{0, .., 9\}$.
We have the following constraints:

$$\begin{aligned}
S + S &= R + 10C_1 && (1)\\
S + D + C_1 &= E + 10C_2 && (2)\\
O + A + C_2 &= G + 10C_3 && (3)\\
R + O + C_3 &= N + 10C_4 && (4)\\
C + R + C_4 &= A + 10C_5 && (5)\\
D &= C_5 && (6)\\
C &\neq 0 && (7)\\
R &\neq 0 && (8)\\
D &\neq 0 && (9)\\
Alldiff(C, R, O, S, A, N, D, G, E) & && (10)
\end{aligned}$$

where $C_1, .., C_5$ are the carry digits with domain $\{0, 1\}$.

Solution

Rule 1: Well you can see that the DANGER has one more letter than CROSS and ROADS, and the extra letter is D. That means that C + R equals something more than 10. Which also means D is 1.

Rule 2: Oh look, S + S = R. That means that R must be even. We have a choice of 4, 6 and 8, because if R was 2, S would have to be 1, and D is already 1. Let's try 6 for the value of R, because we need high numbers if we want C + R to equal something more than 10. Oh look, if R is 6 and S is R divided by 2, then S must be 3!

Rule 3: S+D=E, 3+1=4, So, E=4

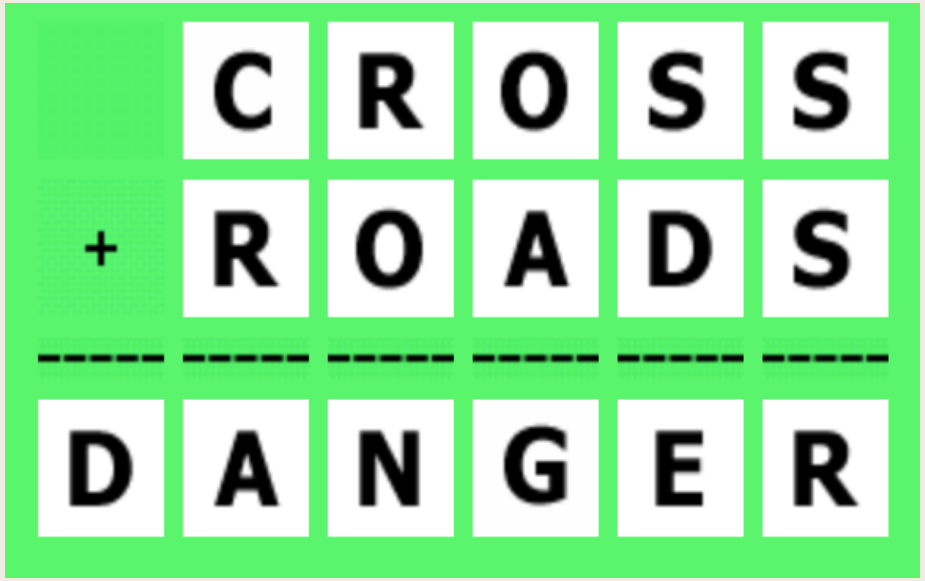
Rule 4: And since we now only have 4 spots in the key left, we choose the highest number for C, which is 9. Again, we need high numbers to make C + R equal something more than 10.

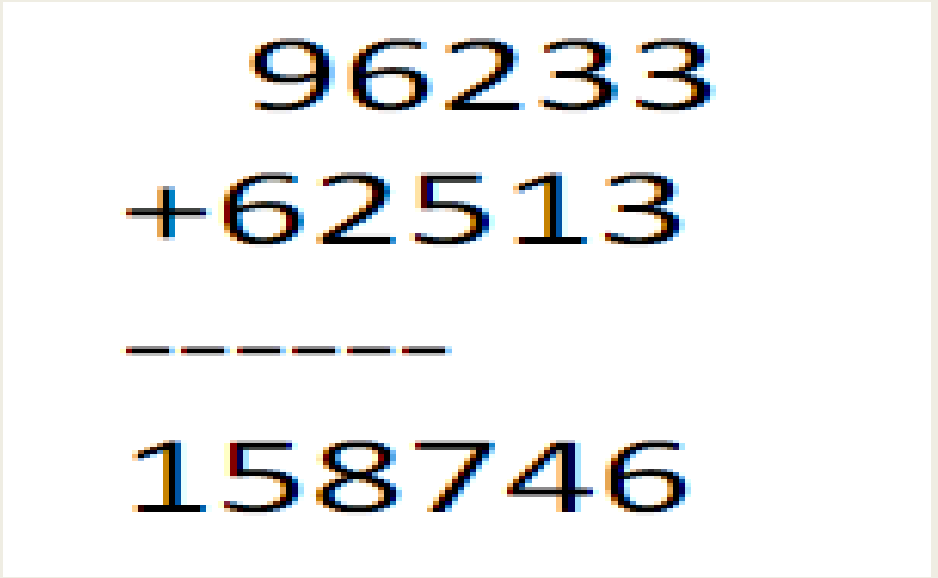
Rule 5: In the equation, O + A = G. We have 2, 5, 7 and 8 vacant. Let's play around with these letters. Let's see if we can find an equation in there. Yes! There is an equation there. 5 + 2 = 7! So G must equal 7. We know that 9 + 6 = 15, but it's missing the 5! So, A must equal 5. In turn, this leads to O having to be 2 (do the maths, O + 5 = 7). And last of all, O is 2, so 6 + 2 (6 + O) = N But 6 + 2 = 8, so now N is 8. We now have the following equation...

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Solution

- Keeping all the constraints in mind, the final resultant is as follows:





	D	O	S	E	A	R	G	N	C
0	1	2	3	4	5	6	7	8	9